

HotSwap Arcade Kit

One mainboard for buttons, joystick, coin acceptor and display. Swap the MCU adapter board to move between ESP32-S3, Arduino NANO, Raspberry Pi Pico 1/2 and Raspberry Pi Zero 2W — no rewiring, no level shifters, no jumper changes.

WHAT CAN YOU BUILD?

- ▶ **Bluetooth arcade controller** — build your own wireless arcade stick for Nintendo Switch, PS5 and PC gaming.
- ▶ **Retro arcade cabinets** — a full control panel for MAME, RetroPie and other retro gaming builds.
- ▶ **Coin-operated machines** — vending projects, claw machines, kiosks and exhibits that take real coins.
- ▶ **Interactive installations** — big-button interfaces for education, museums and maker-fair demos.

WHAT'S IN THE KIT

- 1 Mainboard** — handles all the connectors (buttons, joystick, coin acceptor, display) and powers the whole kit.
- 2 MCU Adapter Board** (choose one) — ESP32-S3, Arduino NANO, Raspberry Pi Pico 1/2, or Raspberry Pi Zero 2W. It plugs straight into the mainboard.

CONNECTOR REFERENCE

CONNECTOR	WHAT IT CONNECTS
Type-C	5V power input for the whole kit
15-pin FPC (1.0mm)	Display OPTIONAL
26-pin header	Mainboard to MCU adapter board
4-pin XH2.54	Coin acceptor
2-pin XH2.54	Buttons and joystick (active-high)

POWER & LOGIC LEVELS

The kit runs entirely from Type-C — no external power adapter needed.

- ▶ Type-C input: **5V**, powers the whole kit
- ▶ Mainboard 5V(OUT) -> Adapter board +5V(IN): powers the MCU
- ▶ Mainboard +12V -> Coin acceptor power
- ▶ Adapter board 3V3(OUT) -> Mainboard V(MCU): logic-level reference

V(MCU) — automatic logic-level switching. Using the reference from the adapter board, the mainboard automatically sets every signal line to 5V or 3.3V logic. No level-shifter modules and no jumper changes when you swap MCUs — just plug in and it works. The adapter board supplies no power; it only provides the reference.

Power indicator: the onboard LED lights up whenever power is present at the Type-C input.

DISPLAY SUPPORT (OPTIONAL)

A display is **not required** — the kit works fully on its own. To show credits, scores or menus, add the **Lonely Binary Display Kit** (sold separately) via the 15-pin FPC connector.

MCU ADAPTER BOARD	DISPLAY SUPPORT
ESP32-S3	Supported
Raspberry Pi Pico 1 / Pico 2	Supported
Raspberry Pi Zero 2W	Supported
Arduino NANO	Not supported (button & coin only)

QUICK START

STEP	ACTION
1	Plug your MCU adapter board into the mainboard's 26-pin header.
2	Connect buttons / joystick (2-pin) and coin acceptor (4-pin).
3	Optionally connect a display via the 15-pin FPC connector.
4	Power the kit through Type-C (5V). The power LED lights up.
5	Flash an example from our GitHub page and start building.

EXAMPLE CODE & FULL DOCUMENTATION



Scan to open our GitHub page

All code, wiring guides and pin maps for every adapter board are here.

github.com/Lonely-Binary/Arcade-HotSwap-Kit